Kidney Transplantation Insights

Outline

- Selection of donor
- Care of donor (deceased)
- Advantage of living donor versus deceased donoer
- Exclusion criteria for living donors
- Preparation of recipient
- Postoperative care of the recipient
- Home discharge instruction

Selection of donor:

- The use of carefully selected donor over 60 years should be maintained and encourage as a continuing source of deceased donor kidney.
- A donor over 70 years should be evaluated on an individual basis taking into account that better result are obtained when transplanted to patients older than 60 years.

Care of donor (deceased):

- Hypertension(catecholamine release)
- Hypovolemia (diabetes insipidus)
- Coagulopathy(hepatic and cardiac dysfunction)
- About 75% of organ donors develop diabetes insipidus due to pituitary necrosis and this leads to hypovolaemia.
- Systemic thermal control is often lost due to hypothalamic ischaemia which results in coagulopathy, hepatic dysfunction and cardiac dysfunction

NOTE:

- Following brain death, a number of physiological changes occur that need to be rectified if donor organ perfusion is to be preserved. After brain death result excess catecholamine release result hypertension and treated with beta blocker, nitroprusside.

- Increased cerebral oedema after trauma or stroke results in catecholamine release .

- With brainstem necrosis, catecholamine levels drop rapidly resulting in hypotension. This should be corrected with fluid and vasopressors.

Advantage of living donor versus deceased donor:

- Graft survival rates for kidneys from living donor is 95% 1 yr and 76% 5 yrs vs graft survival from a cadaveric kidney donor is 89% 1 yr and 61% 5 yrs.
- Better results (both long- and short-term) compared to deceased donor grafts
- Consistent early function and easier management
- Avoidance of long waiting time for transplantation
- Less aggressive immunosuppressive regimens
- Emotional gain to donor
- Increases globally the kidney transplant rate

Exclusion criteria for living donors

1. Absolute contraindications

- Age < 18 years
- Uncontrolled hypertension
- Diabetes mellitus
- Proteinuria (> 300 mg/24 h)
- Abnormal GFR for age

Absolute contraindications

- Microscopic haematuria
- High risk of thromboembolism
- Medically significant illness (chronic lung disease, recent malignant tumour, heart disease)
- History of bilateral kidney stones
- HIV positive

Note:

Active malignancy is a contraindication for transplantation because immunosuppressive therapy may aggravate underlying malignancy. Patients with a history of malignancy should be cured

2. Relative contraindications

- Use of graft with anatomical anomalies
- Active chronic infection (e.g. tuberculosis, hepatitis B/C, parasites)
- Obesity
- Psychiatric disorders

Preparation of recipient:

1. Use of preoperative beta –blocker may be considered in patient at high cardiovascular risk undergoing renal transplant but should be introduced at least 1 month before renal transplant

2. B-blocker should be discontinued suddenly preoperatively

3. Low dose aspirin and plavix therapy are not contraindication to transplant

4. Patient should be strongly encourage to stop smoking and after transplantation

5. Obese patient BMI >30 kg\m2 should be screened vigorously for cardiovascular disease and each case considered individually

Care of recipient:

1. **Immediate postoperative care :goals**

- Maintain of fluid and electrolytes balance
- Wound care
- Pain management(patient suffer from pain caver by pain killer around the clock)
- Good pulmonary hygiene with incentive spirometery, early ambulation and
- Restoration of normal bowel elimination(patient high risk for paralytic illus, constipation caused by steroid and phosphopider)

Postoperative care of the recipient:

- Infection
- CHF
- Hypertension (75%-85%)
- Hyperlipidemia (60%)
- CVS (16%-25%)
- DM (17%- 20%)(more likely to be present before transplantation and new onset DM after transplantation is related to corticosteriod use
- Malignancy affects the skin (40%) or lymphatic system (11%)
- CHF is the most common of hospital admission after renal transplant management of CVD after renal transplant should include modifying factor that control CHF as well as ischemic disease

Annual screening

- Lifelong regular post-transplant follow-up by an experienced and trained transplant specialist is strongly recommended at least every 6-12 months

- Monitoring of renal function and immunosuppression and side-effects by a physician every 4-8 weeks is strongly advise

- Annual screening should include a dermatological examination, tumor screening (including a nodal examination, faecal occult screening, chest X-ray, gynaecological and urological examination), and an abdominal ultrasound, including ultrasound of the native and transplanted kidney)

Annual screening

- Special attention during post-transplant care should also focus on proteinuria, recurrence of original disease

- Post transplant care should aim to detect cardiac disease and cardiovascular risk factors. Cardiac exam and cardiac history should be taken, and if appropriate further diagnostic tests should be prompted to exclude the progression of cardiac disease

- Blood pressure, blood glucose and blood lipids should be determined at appropriate intervals, and adequate measures to control these risk factors should be instituted.

Home discharge instruction:

1. Call your physician or nurse if you have:

- A fever over 100 °F.
- Flu-like symptoms such as chills, aches, headaches, dizziness, nausea or vomiting
- New pain or tenderness around the transplanted kidney
- Fluid retention (swelling)
- Sudden weight gain greater than 2 to 4 pounds within a 24-hour period
- Significant decrease in urine output.

2. Follow recommended dietary plan.

3. Follow recommended fluid intake.

Exercise Guidelines after Kidney Transplant

- In general, to achieve maximum benefits, you should gradually work up to an exercise session lasting 20 to 30 minutes, at least three to four times a week. Exercising every other day will help you keep a regular exercise schedule.

- Wait at least 1½ hours after eating a meal before exercising.

- Dress for the weather conditions and wear protective footwear

- Take time to include a 5-minute warm-up, including stretching exercises, before any aerobic activity and include a 5- to 10-minute cool down after the activity. Stretching can be done while standing or sitting

- Schedule exercise into your daily routine. Plan to exercise at the same time every day (such as in the mornings when you have more energy).

- Add a variety of exercises so that you do not get bored

Exercise precautions

- If changes have been made in your medications, ask your doctor if you can continue your regular exercise program. New medications can affect your response to activity.
- Avoid heavy lifting, pushing heavy objects, and chores such as raking, shoveling, mowing, and scrubbing. When lifting any object, exhale while lifting.
- Avoid even short periods of bed rest after exercise, since it reduces exercise tolerance.

Exercise precautions

- If you develop a rapid or irregular heart beat or has heart palpitations, rest and tries to calm yourself. Check your pulse after resting for 15 minutes. If your pulse is still above 120 to 150 beats per minute, call your doctor for further instructions.

- Do not ignore pain. If you have chest pain or pain anywhere else in your body, do not continue the activity. If you continue to perform an activity while you are in pain, you may cause stress or damage on your joints. Ask your doctor or physical therapist for specific guidelines. Learn to "read" your body and know when you need to stop an activity.

Exercise precautions

Stop exercising and rest if you:

- Have chest pain
- Feel weak, are dizzy or lightheaded
- Have unexplained weight gain or swelling (call your doctor right away)
- Have pressure or pain in your chest, neck, arm, jaw, or shoulder
- Have any other symptoms that cause concern


Thank You